

Bibek Koirala

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PROFESSIONAL SUMMARY

Software engineer and ML practitioner (MS Computer Science, CGPA 4.0/4.0, SIUC) with 5+ years of combined experience building end-to-end ML pipelines, data-centric backend services. Proven ability to design reliable APIs, operate distributed systems, and accelerate ML model delivery. Experienced collaborating cross-functionally with domain experts and product teams to deliver clean, reusable components; former Army Officer with a disciplined, high-accountability work ethic.

TECHNICAL SKILLS

Languages & Query: Python (primary), SQL (PostgreSQL, MySQL), C, C++, Java

ML / AI Frameworks: PyTorch, TensorFlow, scikit-learn, pandas, NumPy, LangChain, LangGraph, HuggingFace

Data & Platforms: PostgreSQL, FAISS (vector store), REST API design, data modelling

MLOps & Cloud: Git, Docker, GitHub Actions (CI/CD), Experiment Tracking, Model Versioning, Linux, Jupyter, VS Code

DL / ML Specialization: Instance Segmentation, Object Detection, Statistics, NLP, RAG, Transformers, LLM, Agentic AI, Regression, Ensemble Methods

RESEARCH & PROFESSIONAL EXPERIENCE

Research Assistant - Machine Learning & Computer Vision

Aug 2024 – Present

Southern Illinois University Carbondale (SIUC)

- Designed and deployed a full **end-to-end instance segmentation pipeline** (Mask R-CNN) on large time-series microscopy datasets - owning data ingestion, model training, and evaluation; achieved mIoU 0.91–0.98 and mAP@0.5 $\approx$ 0.95. Accepted for oral presentation at *IEEE ICMLA 2025*. 🌐 github.com/vbek/bacteria-detection
- Built **SEEKBOT**, a **RAG system** with a 10-node **LangGraph** pipeline, local **FAISS vector store**, and HuggingFace Inference API (Llama 3.1 8B / Qwen 2.5 72B - enabling semantic search over multi-format documents with persistent session state across multi-format documents. 🌐 github.com/vbek/SEEKBOT
- Developed an automated **NLP-based news credibility detection framework** integrating linguistic, semantic, and source-based features; achieved 97% precision and 98% recall with full validation test sets. 🌐 github.com/vbek/News-Credibility
- Performed **comparative deep-learning model analysis** (CNN, fine-tuned ResNet50, vanilla ResNet, ViT) on the PlantVillage dataset; trained and validated achieving 97.9–99.6% accuracy across 38 disease classes. 🌐 github.com/vbek/Plant-Disease-Detection
- Conducted **EDA and ML clustering** on macroeconomic financial datasets (1960–2023) to identify patterns in high-dimensional indicators; applied econometric modelling to validate findings. 🌐 github.com/vbek/Inflation-Dynamics
- Analyzed a sparse, high-dimensional criminology survey dataset to identify victimization patterns and cluster victims by reporting behavior; results aligned strongly with prior research benchmarks. 🌐 github.com/vbek/Victim-Crime-Perception

Officer - Software Development & R&D

May 2018 – Dec 2023

Nepali Army

- Designed and **deployed a Windows-based GUI dashboard with a relational database backend** to manage troop records enabling a fully paperless operational workflow. 🌐 github.com/vbek/Military-Leave-Management-Dashboard
- Engineered **automated artillery guidance software** with optimized numerical computation pipelines - reduced targeting error from 50 m to 10 m and computation latency from minutes to microseconds.
- Developed **decision-support software** for logistical calculations under real-world operational constraints; collaborated with cross-functional teams to ensure system reliability, accuracy, and operational readiness.
- Demonstrated critical thinking, resilience, and ethical decision-making under pressure - core competencies for fast-moving, ambiguous distributed systems environments.

Data Analyst

Dec 2016 – Feb 2018

Morang Hospital Pvt. Ltd.

- Managed and maintained **PostgreSQL relational databases** for inventory and patient records; wrote **SQL queries** to extract, join, and aggregate large financial transaction and procurement datasets.
- Performed **EDA on patient admissions, billing, and inventory datasets**; produced dashboards and operational reports communicating data-driven insights to non-technical stakeholders.
- Tracked pharmaceutical procurement and sales transactions, identifying discrepancies and financial inefficiencies, demonstrating a data quality and operational excellence mindset.

EDUCATION

Southern Illinois University Carbondale, MS Computer Science

Aug 2024 – Present

Research Assistant (Computer Vision | Object Detection)

- CGPA: 4.0/4.0
- **Coursework:** Machine Learning & Soft Computing, Deep Learning, Data Mining, Natural Language Processing, Artificial Intelligence, Parallel & Distributed Computing, Advanced Database Systems, Cryptography & Network Security

Pulchowk Campus, IOE, Tribhuvan University, BE Computer Engineering

Nov 2012 – Sep 2016

Top 1% of 15,000 applicants - Full Scholarship Recipient

- **Coursework:** Big Data, AI, Database Systems, Data Structures & Algorithms, Signal Processing, OS, Software Development, Computer Security, Assembly Language, Computer Architecture

LEADERSHIP & ADDITIONAL

Former Army Officer with demonstrated leadership under pressure, cross-functional team collaboration, and high-stakes decision-making in ambiguous environments. Actively participates in knowledge-sharing and peer collaboration within technical teams. Captained Pulchowk Campus cricket team (2015–2016); district-level cricket player (2016–2017). Also active in volleyball, chess, badminton, and running.